ONLINE SALES ANALYSIS USING POWER BI

1. Executive Summary

- This project analyzes online sales transaction data to evaluate business performance across products, regions, and time. Using Power BI, the dataset was transformed, modeled, and analyzed to identify revenue drivers, profitability trends, and operational inefficiencies.
- The objective was not just visualization — but extracting meaningful, decision-support insights from raw transactional data.

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2. Analytical Approach

- The project followed a structured data analytics workflow:
- Data Cleaning and Validation
- Data Modeling using Star Schema
- KPI Definition using DAX
- Exploratory Data Analysis (EDA)
- Trend and Performance Evaluation
- Business Insight Extraction
- This ensures analytical integrity rather than surface-level dashboard building.

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3. Data Preparation and Validation

- Before analysis, the dataset was examined for:
- Missing values

- Duplicate transactions
- Incorrect data types
- Inconsistent category labels
- Outlier sales values
- Time intelligence columns (Year, Month, Quarter) were created to support trend analysis.
- This stage ensured data reliability and prevented aggregation distortions.

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4. Key Performance Indicators Defined

- To evaluate performance objectively, the following KPIs were defined:
- Total Revenue
- Total Profit
- Total Quantity Sold
- Profit Margin (%)
- Year-over-Year Growth
- Average Order Value
- Each KPI was calculated using DAX measures to ensure dynamic filtering capability.

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5. Analytical Findings

5.1 Revenue Concentration

- Analysis shows that revenue distribution is not uniform across products. A small percentage of products contribute to a disproportionately high share of total revenue. This follows the Pareto Principle (80/20 rule).

- Implication: Business performance depends heavily on key products.

5.2 Profitability Variation

High revenue does not necessarily imply high profitability. Certain products generate strong sales volume but lower margins, possibly due to:

- Heavy discounting
- High logistics cost
- Supplier pricing

Implication: Revenue optimization must be aligned with margin strategy.

5.3 Regional Performance Gap

Regional analysis indicates variation in:

- Revenue contribution
- Profit margins
- Growth rate

Some regions demonstrate strong revenue but weaker profitability, indicating operational inefficiencies.

Implication: Regional strategy must focus on cost optimization and pricing control.

5.4 Seasonal Trends

Monthly trend analysis reveals fluctuations in sales volume across different months. Certain periods show consistent spikes, suggesting:

- Seasonal demand

- Promotional campaigns
- Festive impact

Implication: Forecasting models should incorporate seasonal patterns.

5.5 Customer Contribution Pattern

Sales distribution suggests dependency on a limited customer base contributing a significant revenue share.

Implication: Customer retention strategies are critical for revenue stability.

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6. Risk Analysis

Potential business risks identified:

- Over-dependence on limited product categories
- Margin erosion due to uncontrolled discounting
- Regional imbalance in performance
- Revenue volatility due to seasonality

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7. Strategic Recommendations

1. Focus marketing investment on high-margin products.
2. Re-evaluate pricing strategy for low-margin high-volume items.
3. Strengthen supply chain efficiency in low-profit regions.
4. Develop predictive forecasting using seasonal data.
5. Implement loyalty programs for top customers.

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8. Technical Evaluation

The project demonstrates:

- Structured data modeling (Star Schema)
- Advanced DAX calculation usage
- Time intelligence implementation
- Interactive filtering logic
- Business-focused dashboard design

The analytical depth goes beyond descriptive statistics and supports prescriptive decision-making.

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9. Limitations

- Analysis is dependent on historical data only.
- No external economic factors included.
- Customer behavioral segmentation not deeply explored.
- Predictive modeling not implemented.

Future improvements can include machine learning-based sales forecasting.

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10. Conclusion

- This project successfully transformed raw sales data into structured business intelligence. The analysis reveals revenue concentration, profitability imbalance, regional variation, and seasonal patterns.
- The study highlights the importance of aligning revenue growth with profitability control and strategic decision-making.